Report

Exercise 3 - MongoDB

Group: 82
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Introduction

I completed the exercise individually using The Movies Dataset. The work involved exploratory data analysis, data cleaning, document design, data import, and analytical queries using Python and MongoDB. MongoDB ran locally in Docker, and Git was used for version control.

Repository: https://github.com/verinix/tdt4225-exercise-3

Results

Part 1 - EDA, data cleaning, and document design

I began by inspecting the main distributions and data-quality issues before deciding how to clean and store the data.

Raw dataset	Result
Movie metadata rows	45,466
Credits rows	45,476
Keywords rows	46,419
Links rows	45,843
Ratings	26,024,289
Distinct rating users	270,896
Invalid movie IDs	3
Duplicated metadata IDs	29
Duplicated credits IDs	43
Duplicated keywords IDs	985

Ratings greatly outnumbered movies, with an average of 96.07 ratings per user (median of 30). The most active user submitted 18,276 ratings, while ratings ranged from 0.5 to 5.0. The movie metadata contained significant gaps in financial details; only 7,408 movies reported positive revenue and 8,890 reported positive budgets. The median revenue was approximately $16.82 million, with a median runtime of 95 minutes. Additionally, 2,442 movies lacked genres and 87 were missing release dates.

```
Credits
Rows: 45,476
Distinct movie IDs: 45,432
Invalid movie IDs: 0
Duplicated movie IDs: 43
Malformed cast/crew rows: 0
Identical duplicates: 36
Conflicting duplicates: 7

Cast and crew
Cast: empty=2,418, mean=12.37, median=10, max=313
Crew: empty=771, mean=10.21, median=6, max=435
Movies without a director: 887
Movies with multiple directors: 3,200
Movies with repeated cast person IDs: 953
Movies with repeated cast orders: 87

Cast gender values
0: 223,964
1: 111,797
2: 226,713

Keywords
Rows: 46,419
Distinct movie IDs: 45,432
Invalid movie IDs: 0
Duplicated movie IDs: 985
Malformed keyword rows: 0
Identical duplicates: 985
Conflicting duplicates: 0

Keyword counts
Empty lists: 14,795
Maximum: 149
Invalid/missing keyword names: 4

Links
Rows: 45,843
Distinct MovieLens IDs: 45,843
Invalid MovieLens IDs: 0
Missing TMDB IDs: 219
Invalid TMDB IDs: 0
Distinct mapped TMDB IDs: 45,594
TMDB IDs shared by multiple MovieLens IDs: 29
```

```
Movie metadata
Rows: 45,466
Distinct movie IDs: 45,433
Invalid movie IDs: 3
Duplicated movie IDs: 29
Identical duplicates: 16
Conflicting duplicates: 13

Missing text fields
tagline: 25,052
overview: 959
original_language: 11
title: 3

Missing release dates: 87
Invalid release dates: 0

Numeric fields
budget: missing=0, zero=36,573, invalid=0, positive=8,890
revenue: missing=3, zero=38,052, invalid=0, positive=7,408
runtime: missing=260, zero=1,558, invalid=0, positive=43,645
vote_average: missing=3, zero=2,998, invalid=0, positive=42,462
vote_count: missing=3, zero=2,899, invalid=0, positive=42,561
Median positive revenue: $16,822,718
Median positive runtime: 95.0 minutes

Nested fields
genres: missing=0, empty=2,442, malformed=0
production_companies: missing=3, empty=11,875, malformed=0
production_countries: missing=3, empty=6,282, malformed=0
Collections: present=4,491, missing=40,972, malformed=0

Movies by decade
1870s: 2
1880s: 4
1890s: 75
1900s: 87
1910s: 176
1920s: 432
1930s: 1,317
1940s: 1,494
1950s: 2,080
1960s: 2,622
1970s: 3,472
1980s: 3,931
1990s: 5,677
2000s: 11,207
2010s: 12,799
2020s: 1
```

```
Ratings
Rows: 26,024,289
Invalid rows: 0
Valid rows: 26,024,289
Distinct users: 270,896
Distinct rated movies: 45,115

Rating distribution
0.5: 404,897
1.0: 843,310
1.5: 403,607
2.0: 1,762,440
2.5: 1,255,358
3.0: 5,256,722
3.5: 3,116,213
4.0: 6,998,802
4.5: 2,170,441
5.0: 3,812,499

Ratings per user
Mean: 96.07
Median: 30
Maximum: 18,276
Users with >= 20 ratings: 166,444

Movie metadata relationships
Credits: matched=45,432, metadata without credits=1, credits without metadata=0
Keywords: matched=45,432, metadata without keywords=1, keywords without metadata=0
Movies present in all three files: 45,432

TMDB mapping coverage
Linked TMDB IDs found in metadata: 45,433
Linked TMDB IDs absent from metadata: 161
Metadata IDs absent from links: 0

Ratings coverage
Matched: movies=44,741, ratings=25,981,582
No MovieLens link: movies=0, ratings=0
Missing TMDB ID: movies=213, ratings=13,503
Missing movie metadata: movies=161, ratings=29,204
```

Analysis of duplicates separated identical entries from conflicting ones. Of the 29 duplicated metadata IDs, 16 were identical and 13 were conflicting. Credits contained 36 identical and 7 conflicting IDs, while all 985 duplicated keyword IDs were identical.

The files were largely consistent through their movie identifiers. About 99.84% of ratings could be linked to movies with metadata, while the remaining ratings included 13,503 without usable TMDB links and 29,204 entries where the linked movie metadata was missing.

Cleaning

I kept one copy of identical duplicates and discarded any conflicting records, as choosing between them would have been arbitrary. I also filtered out invalid movie IDs and keywords without valid names.

Missing text and dates were stored as “null”. Budgets, revenues, and runtimes with zero or missing values were treated as unavailable, though zero vote averages were kept. Movies with partial attributes were preserved because they could still be utilized for other analyses.

Collections

I organized the data into two collections: movies and ratings.

The movies collection contains one document per TMDB movie, embedding attributes like genres, cast, crew, keywords, production companies, production countries, and collection. These fields were grouped because they are linked to unique movies and are typically accessed together.

The ratings collection stores each user rating, including user ID, MovieLens and TMDB identifiers, rating, and timestamps. I opted not to embed ratings within movie documents to avoid creating large, uneven documents.

TMDB IDs serve as the primary movie identifiers. Dates and numerical values were stored using native MongoDB types. I used batched inserts and created a text index on overview and tagline to support the search task. Some metadata fields, such as original_title, spoken_languages, and status, were omitted because they were not needed for the analytical queries.

Import result

Collection	Documents
movies	45,417
ratings	26,024,289

The import preserved all ratings but removed invalid or conflicting movie records. Ratings without metadata remained available for user-based statistics.

```
Duplicate cleaning
movies_metadata.csv: 29 duplicated, 16 identical, 13 conflicting IDs
credits.csv: 43 duplicated, 36 identical, 7 conflicting IDs
keywords.csv: 985 duplicated, 985 identical, 0 conflicting IDs
Excluded movie IDs: 16
Movies inserted: 45,417
Ratings inserted: 26,024,289
```

Part 2 - Analytical queries

Missing revenues and runtimes were excluded from their respective statistics, while the movies could still contribute to movie counts. Repeated actor or director IDs were counted once per movie when counting movie credits and collaborations. For large result sets, only 20 rows are displayed, although the complete result sets were calculated. Unknown genders were excluded from the cast proportions. For Task 9, I used recorded US production countries because company nationalities were unavailable.

Task 1

Kyle Balda recorded the highest median revenue among directors with at least five movies, at approximately $1.02 billion, followed by David Yates and Peter Jackson.

```
Task 1
Director           Movies    Median revenue ($)    Mean vote average
---------------  --------  --------------------  -------------------
Kyle Balda              5         1020063384.00                6.62
David Yates            10          936085967.50                6.03
Peter Jackson          14          871368364.00                7.12
Joss Whedon             5          722136579.00                7.40
George Lucas            7          712398167.50                6.81
Chris Renaud            5          709485961.00                6.70
Mike Thurmeier          6          642911910.00                6.67
Brad Bird               6          623722818.00                7.15
Francis Lawrence        6          619388635.50                6.80
Raman Hui               5          592121491.00                6.22
```

Task 2

I identified 45,641 actor pairs that appeared together in at least three movies. Huntz Hall and Leo Gorcey had the most (35), followed by Charlie Chaplin and Edna Purviance (33).

```
Task 2
Qualifying pairs: 45,641
Actor 1          Actor 2             Movies    Mean vote average
---------------  ---------------  --------  -------------------
Huntz Hall       Leo Gorcey             35                3.63
Charlie Chaplin  Edna Purviance         33                6.46
Oliver Hardy     Stan Laurel            31                6.35
Jeff Bennett     Rob Paulsen            27                6.23
Lou Costello     Bud Abbott             27                6.47
Grey Griffin     Frank Welker           25                6.71
Raymond Burr     Barbara Hale           25                4.98
John Wayne       Paul Fix               24                5.85
Jim Cummings     Frank Welker           21                6.55
Frank Welker     Jeff Bennett           21                6.23
Frank Mayo       Jack Mower             21                5.39
Peter Cushing    Christopher Lee        21                5.99
Jim Cummings     Jeff Bennett           21                6.27
Charlie Chaplin  Henry Bergman          20                6.98
Bernard Gorcey   Leo Gorcey             20                4.55
Toshirō Mifune   Takashi Shimura        19                7.48
Buster Keaton    Joe Roberts            19                6.88
Charlie Chaplin  John Rand              19                6.75
Jackie Chan      Yuen Biao              19                6.80
Simon Yam        Lam Suet               19                6.10
Showing first 20 pairs.
```

Task 3

Christopher Lee and several other actors appeared in all 20 genres.

```
Task 3
Actor             Movies    Distinct genres  Example genres
---------------  --------  -----------------  ------------------------------------------------
Christopher Lee       148                 20  Adventure, Comedy, Family, Fantasy, Action
David Carradine        60                 20  Action, Crime, Drama, Thriller, Adventure
Michael Madsen         65                 20  Action, Crime, Thriller, Adventure, Family
Jonathan Pryce         55                 20  Adventure, Fantasy, Action, Comedy, Science Fiction
Jim Broadbent          78                 20  Comedy, Science Fiction, Adventure, Action, Family
Stacy Keach            59                 20  Drama, Crime, Thriller, Comedy, Horror
Eddie Izzard           43                 20  Thriller, Crime, Drama, Music, History
Billy Zane             64                 20  Adventure, Comedy, Science Fiction, Family, Drama
Beau Bridges           60                 20  War, Crime, Drama, Mystery, Romance
Bill Nighy             61                 20  Adventure, Fantasy, Action, Thriller, Comedy
```

Task 4

The Harry Potter Collection generated the most revenue (roughly $7.71 billion), followed by Star Wars ($7.43 billion) and James Bond ($7.11 billion).

Task 4

Collection	Movies	Total revenue ($)	Median vote average	Earliest release	Latest release
Harry Potter Collection	8	7707367425.00	7.50	2001-11-16	2011-07-07
Star Wars Collection	8	7434494790.00	7.45	1977-05-25	2016-12-14
James Bond Collection	26	7106970239.00	6.30	1962-10-04	2015-10-26
The Fast and the Furious Collection	8	5125098793.00	6.65	2001-06-22	2017-04-12
Pirates of the Caribbean Collection	5	4521576826.00	6.90	2003-07-09	2017-05-23
Transformers Collection	5	4366101244.00	6.10	2007-06-27	2017-06-21
Despicable Me Collection	6	3691070216.00	6.90	2010-07-08	2017-06-15
The Twilight Collection	5	3342107290.00	5.80	2008-11-20	2012-11-13
Ice Age Collection	5	3216708553.00	6.50	2002-03-10	2016-06-23
Jurassic Park Collection	4	3031484143.00	6.35	1993-06-11	2015-06-09

Task 5

Early movies typically had shorter runtimes, whereas median runtimes for various genres in more recent decades averaged between 90 and 110 minutes.

Task 5

Groups found: 232

Decade	Primary genre	Movies	Median runtime (min)
2010	Crime	319	103.0
2010	Drama	3184	100.0
2010	Action	1048	100.0
2010	War	75	100.0
2010	Romance	392	99.0
2010	Music	121	99.0
2010	Western	28	98.5
2010	Adventure	303	97.0
2010	Foreign	28	96.5
2010	Mystery	157	95.5
2010	Thriller	778	95.0
2010	Fantasy	143	93.0
2010	Comedy	2323	93.0
2010	Science Fiction	185	91.0
2010	Horror	784	90.0
2010	Family	203	90.0
2010	Documentary	1626	89.0
2010	TV Movie	149	87.0
2010	Animation	327	80.0
2020	Action	1	

Showing last 20 groups (most recent decades).

Task 6

The proportion of women among the top five billed cast members rose from 0.311 in the 1970s to 0.376 by the 2010s.

Task 6

Decade	Mean female proportion	Movies
1870	1.000	1
2020	0.800	1
1890	0.500	2
1900	0.444	12
2010	0.376	10538
1930	0.366	1247
2000	0.361	9643
1940	0.359	1434
1990	0.338	5177
1910	0.337	144
1950	0.336	1997
1920	0.331	374
1960	0.325	2395
1980	0.320	3554
1970	0.311	3145

Task 7

The text search identified five movies mentioning noir in their overview or tagline with at least 50 votes. The Bad Sleep Well and Drunken Angel had the highest vote averages, both at 7.70.

Task 7

Title	Year	Vote average	Vote count
The Bad Sleep Well	1960	7.70	57
Drunken Angel	1948	7.70	54
Elevator to the Gallows	1958	7.60	85
Frank & Lola	2016	5.70	51
Synchronicity	2015	5.70	114

Task 8

I found 1,390 pairs with three or more collaborations among movies with at least 100 votes. Leading pairs included Charlie Chaplin and Hank Mann, and several Akira Kurosawa collaborations, with mean vote averages of approximately 8.13.

Task 8
Qualifying pairs: 1,390

Director	Actor	Movies	Mean vote average	Mean revenue ($)
Charlie Chaplin	Hank Mann	3	8.13	6506393.67
Akira Kurosawa	Eijirô Tôno	3	8.13	271841.00
Akira Kurosawa	Minoru Itô	3	8.13	271841.00
Akira Kurosawa	Haruo Suzuki	3	8.13	271841.00
Quentin Tarantino	Harvey Keitel	3	8.10	182573606.33
Francis Ford Coppola	John Cazale	3	8.10	99009750.67
Akira Kurosawa	Atsushi Watanabe	3	8.10	163540.50
Akira Kurosawa	Toranosuke Ogawa	3	8.10	163540.50
Francis Ford Coppola	Robert Duvall	4	8.07	96622408.25
Akira Kurosawa	Daisuke Katô	4	8.07	141216.33
Akira Kurosawa	Ichirô Chiba	4	8.07	163540.50
Akira Kurosawa	Noriko Honma	4	8.07	141216.33
Akira Kurosawa	Haruya Sakamoto	3	8.07	271841.00
Akira Kurosawa	Shôichi Hirose	3	8.07	271841.00
Akira Kurosawa	Susumu Fujita	3	8.07	
Akira Kurosawa	Isao Kimura	4	8.05	163540.50
Akira Kurosawa	Minoru Chiaki	6	8.03	141216.33
Akira Kurosawa	Masao Shimizu	3	8.03	55240.00
Akira Kurosawa	Yûnosuke Itô	3	8.03	55240.00
Hayao Miyazaki	Tsunehiko Kamijô	3	8.03	217150201.50

Showing first 20 pairs.

Task 9

For movies produced in the US, French was the most common non-English language (112 movies), followed by Spanish (72), Italian (56), and German (51).

Task 9

Original language	Movies	Example title
fr	112	Wings of Courage
es	72	Bitter Sugar
it	56	Frankie Starlight
de	51	Cold Fever
ja	29	Godzilla 1985
pt	15	Senseless
xx	14	Quest for Fire
nl	12	Come On, Rangers
ru	11	Dark Eyes
zh	11	Eat Drink Man Woman

Task 10

The most diverse users had rated movies in all genres. The highest population rating variances approached 5.0625, the maximum possible for ratings between 0.5 and 5.0, indicating highly polarized ratings.

```
Task 10: Most diverse
  User    Ratings    Population variance    Distinct genres
------  ---------  ---------------------  -----------------
   229       1282                  1.177                 20
   231       1352                  0.411                 20
   567        972                  1.299                 20
   741       2811                  0.470                 20
   836       1521                  0.500                 20
   924        969                  1.004                 20
  1104       1057                  0.721                 20
  1145        553                  1.328                 20
  1214        165                  0.765                 20
  1380       1122                  0.378                 20

Task 10: Highest rating variance
  User    Ratings    Population variance    Distinct genres
------  ---------  ---------------------  -----------------
 18703         20                  5.062                 13
 57842         20                  5.062                 14
 83102         20                  5.062                 15
153882         20                  5.062                 15
167241         34                  5.062                 14
185889         26                  5.062                 14
 15395         83                  5.062                 14
 71009         27                  5.056                 13
179759         21                  5.051                 13
221535        120                  5.040                 19
```

Discussion

The primary cleaning decision involved handling duplicate movie IDs. I kept one instance of identical entries but removed conflicting ones to avoid arbitrary selection. Movies with partial data were kept, as missing values in one category did not render the remaining data useless for analysis.

The document design was shaped by the size and relationships of the dataset. I embedded cast, crew, and genres within movie documents to simplify queries for related attributes. Conversely, I stored ratings separately due to their large volume.

The main analytical limitation was due to incomplete metadata, specifically regarding financial data. Missing revenue figures impacted director, collection, and collaboration statistics, while missing runtimes and genres affected other areas. Other constraints included the limited number of genres and data for early decades, and the reliance on specific wording for text searches.

Feedback

No specific feedback.

AI use

I used AI to assist with understanding MongoDB functionality and writing aggregation pipelines. I also used AI to review the implementation and submission for correctness.